

Thesis Proposal Defence — Panel Comments & Questions

Title: Multi-Agent System for Automated Cardiac MRI Analysis and Decision Support

PANEL 1 COMMENTS: Dr. Burhan Ul Islam Khan

C1. Citation Years Are Systematically Off: In-text citation years do not match the reference list for multiple papers including Salih (cited 2025, listed 2023), Baba (cited 2025, listed 2026), Sahoo (cited 2025, listed 2024), Zeng (cited 2025, listed 2024), and Bennai. All in-text citations must be verified and corrected to match their reference list entries exactly.

C2. Dataset Inconsistency Across Sections: The abstract, scope (Table 1.2), and Objective O4 all commit to two datasets, ACDC and M&Ms (525 subjects = 150 + 375). However, the methodology introduces a third dataset (M&Ms-2) without acknowledging this addition in the abstract, scope, or objectives. This must be made consistent throughout the proposal.

C3. RV Analysis Scope Contradiction: Table 1.2 lists RV (Right Ventricular) analysis as out of scope. Yet the Segmentation Agent (Section 3.3.1) produces RV cavity masks, and the expected outcomes promise left ventricular and RV results. This is a direct internal contradiction that must be resolved, either include RV analysis in scope or remove it from the methodology.

C4. Baseline Count Mismatch: Section 3.8 states "five" baselines but Table 3.4 lists only four. The number must be made consistent and all baselines must be explicitly named and accounted for.

C5. Chapter and Table Numbering Broken: Chapter 3 numbering is broken, sections skip 3.9 and 3.10, and tables skip 3.2 and 3.3. A full structural review of Chapter 3 is needed to correct all numbering sequencing.

C6. Reference List Not Alphabetically Ordered: The reference list must be arranged alphabetically letter by letter by the surname of the first author. The current list is unordered and does not comply with APA formatting requirements.

C7. Ambiguous and Malformed Citations: "Fu et al., 2025" and "Zhang et al., 2025" each point to two different papers and must be disambiguated as 2025a and 2025b. "Kristijan et al." is malformed and needs to be corrected with the full proper APA citation.

C8. BERTScore Mislabelled: Section 2.4.2 incorrectly groups BERTScore with BLEU and ROUGE as a "lexical" metric. BERTScore is embedding-based and semantic, not lexical. The classification and description must be corrected.

PANEL 2 COMMENTS: Dr. Mohamad Hazim Md Hanif

C9. Use of Private/Proprietary Dataset Since this is a Master by Research programme, the candidate is expected to collect and process their own data. The panel strongly recommends

incorporating a proprietary hospital dataset to ensure the candidate undergoes the full data collection and preprocessing process, which is a core research requirement.

C10. Error Propagation in Multi-Agent Pipeline: If the nnU-Net segmentation agent commits an error on a difficult apical slice, how will the framework prevent this error from propagating downstream and causing cascading failures in quantification, clinical reasoning, and explainability? A concrete error containment and propagation mitigation strategy must be described.

C11. Conflicting Clinical Guidelines: How will the system handle instances where clinical guidelines from the ESC and ACC/AHA present conflicting recommendations or different classification cutoffs for the same condition? A conflict resolution mechanism must be explicitly addressed in the methodology.

C12. Iterative Feedback Loop Trigger Not Specified: The precise programmatic trigger for the iterative feedback loop between the clinical reasoning agent and the segmentation agent is not defined. The candidate must specify exactly what condition activates the loop, how it is executed technically, and what the stopping criterion is.

C13. Evaluation Metric Gap: Given that BERTScore only measures semantic similarity and Grad-CAM evaluates network attention, neither directly measures the clinical correctness of the system's diagnostic outputs. The candidate must identify a specific quantitative metric or human-in-the-loop evaluation method that directly assesses diagnostic accuracy and clinical validity.

C14. No AI Training — Be Explicit: The candidate must clearly and explicitly state which components of the system involve no AI training, for example, the quantification agent and orchestration layer, to distinguish trained components from rule-based or retrieval-based ones.

C15. Agent-to-Agent Security: Agent-to-agent security must be a core part of the multi-agent workflow design. The candidate must address how the system protects against malicious prompts, prompt injection between agents, and hallucinated outputs. Guardrails for hallucination mitigation must be explicitly incorporated and described.

C16. Engineering vs. Research Framing: The work is currently framed as a complex engineering integration project rather than a research contribution. The candidate must clearly articulate the novel research contribution, theoretical advancement, and scientific knowledge generated, distinguishing it from a software development or systems integration exercise.

These comments are to be reviewed and addressed in the revised proposal and subsequent thesis chapters.